Course Code : ECT 3002 **UVXV/MW – 24 / 1524**

Third Semester B. Tech. (Electronics and Communication Engineering) Examination

ANALOG CIRCUITS DESIGN

Time : 3 Hours] [Max. Marks : 50

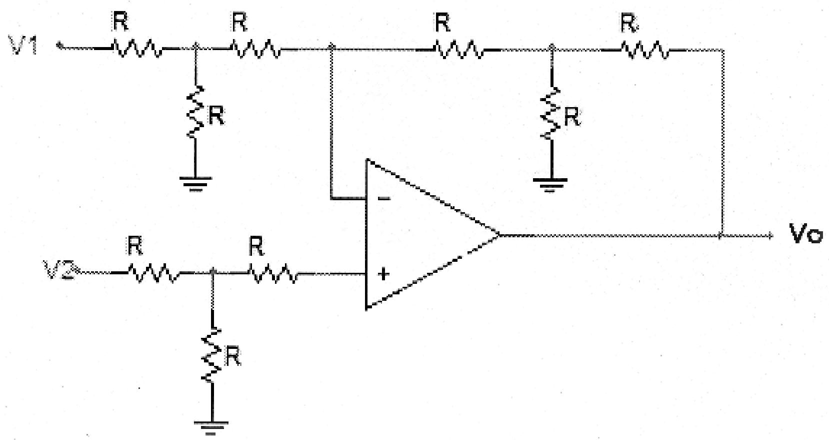
Instructions to Candidates :—

(1) All questions are compulsory.
(2) All questions carry marks as indicated against them.
(3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) Derive the DC equations for Dual input balanced output differential amplifier. 5(CO1)

(b) Elaborate the importance of Level shifter circuit in Differential amplifier. 5(CO1)

2. (a) For the circuit shown in Fig. 2, find V_o in terms of V_1 and V_2.



5(CO2)

(b) Design a practical differentiator circuit to differentiate a sine wave signal of 1 sin (628.31)t. Sketch the input-output waveform. 5(CO2,4)

3. (a) Design a R-C phase shift oscillator using op-amp to oscillate at 500 Hz. 5(CO2,3,4)

(b) Design a second order active high pass Butterworth filter with lower cut-off frequency 10 KHz. 5(CO5)

4. (a) For Schmitt trigger circuit using op-amp, evaluate threshold voltages V_{UT}, V_{LT} hysteresis voltage, and centre of hystersis curve. Assume op-amp with saturation voltage ±12 V, $R_1 = 2\ K\Omega$, $R_2 = 4\ K\Omega$, Vref = –4 V. 5(CO2,3)

(b) Explain the operation of Sample and hold circuits. 5(CO2)

5. (a) Design an astable multivibrator using IC555 having output frequency 5 KHz and 60% duty cycle. 5(CO5)

(b) Illustrate block diagram of Phase lock loop and explain its working in brief. 5(CO2)